

Longest cycles meet linearly in highly connected graphs

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Abstract

In this talk, we present the first linear lower bound toward a conjecture of Smith (1979).

The proof relies on three ingredients:

- a weak-2-connectivity covering theorem, analogous to the block-cut decomposition.
- a generalization of the Bondy–Locke theorem and its application.
- a strengthened Dirac theorem for highly connected graphs.

Smith's conjecture

This work is motivated by the following conjecture.

Conjecture (Smith, 1979)

For every integer $k \geq 2$, any two longest cycles in a k -connected graph share at least k vertices.

Throughout, the connectivity refers to vertex-connectivity.

Sharp example: $K_{k,n-k}$ with $n \geq 3k$. There exist two longest cycles whose intersection is exactly the k vertices in the smaller part.

Smith's conjecture

Conjecture (Smith, 1979)

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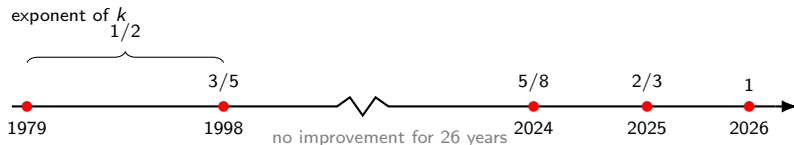
- $k \leq 6$: Grötschel (1984).
- $k = 7, 8$: Stewart–Thompson (1995).
- $k \geq (n+16)/7$: Gutiérrez–Valqui (2024).

A quantitative problem

Quantitative form

Find the largest function $f(k)$ such that any two longest cycles in every k -connected graph share at least $f(k)$ vertices.

- Burr–Zamfirescu (reported in 1998): $f(k) \geq \sqrt{k} - 1$.
- Chen–Faudree–Gould (1998): $f(k) = \Omega(k^{3/5})$.
- Groenland–Longbrake–Steiner–Turcotte–Yepremyan (2024): $f(k) = \Omega(k^{5/8})$.
- Ma–Z. (2025, preprint): $f(k) = \Omega(k^{2/3})$.
- Ma–Ning–Z. (2026, in preparation): $f(k) \geq k/4000$.



Main theorem

Theorem (Ma–Ning–Z.)

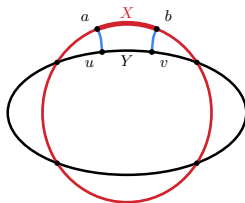
For every integer $k \geq 2$, any two longest cycles in a k -connected graph share at least $k/4000$ vertices.

- The order in k is optimal; the constant is not optimized.
- The proof is based on a covering theorem analogous to decomposing a connected graph into its blocks.

The previous unified idea

Fact

There are no two such disjoint (X, Y) -paths.



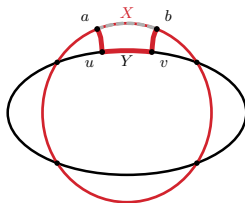
Unified principle in the previous approach:

a forbidden subgraph in an auxiliary graph
 $\Downarrow$ Turán-type extremal bound
 an upper bound on the number of disjoint (X, Y) -paths

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A different approach

Theorem (Ma–Ning–Z.)

For every integer $k \geq 2$, any two longest cycles in a k -connected graph share at least $k/4000$ vertices.

The proof idea is entirely different, as seen in the following model case.

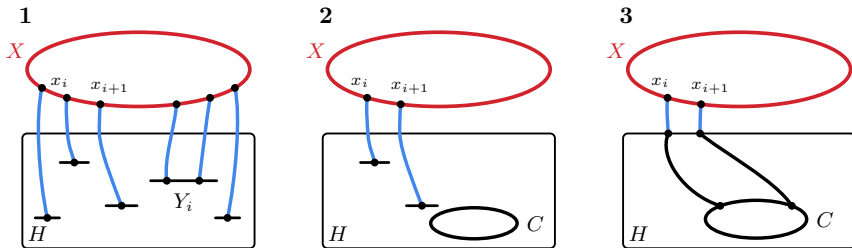
Model case

Let X be a longest cycle. Assume that every component of $G - X$ is 3-connected and has at least k vertices.

Proof idea in the model case

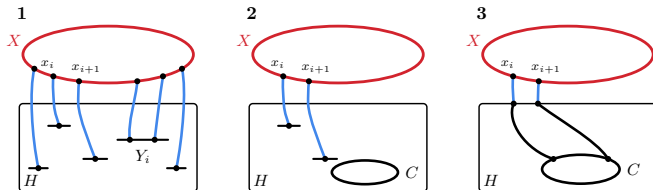
Theorem (Model case)

Let X, Y be longest cycles in a k -connected graph and let $m = |V(X) \cap V(Y)|$. If every component of $G - X$ is 3-connected and has at least k vertices, then $m = \Omega(k)$.



Proof idea in the model case

Write $p(H)$ and $c(H)$ for the number of edges in a longest path and a longest cycle in H , respectively.



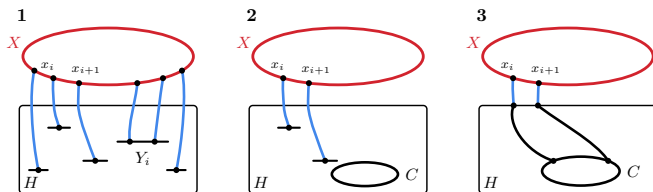
- 1 Choose H to be the component of $G - X$ containing the longest subpath among $Y - X$. Menger gives k disjoint (X, H) -paths.
- 2 Take two paths with consecutive endpoints x_i, x_{i+1} on X . Since $p(H) \geq c(G)/m - 2$, Bondy-Locke gives a cycle $C \subseteq H$ with $|C| = \Omega(p(H)) = \Omega(c(G)/m)$.

Theorem (Bondy-Locke)

Every 3-connected graph H satisfies $c(H) \geq \frac{2}{5}p(H)$.

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- 3 By 2-connectivity, reroute the two paths disjointly to C . The endpoints of these (X, C) -paths on X satisfy

$$\text{dist}_X(x_i, x_{i+1}) \geq |C|/2.$$

Summing over the endpoints on X of the k disjoint (X, H) -paths,

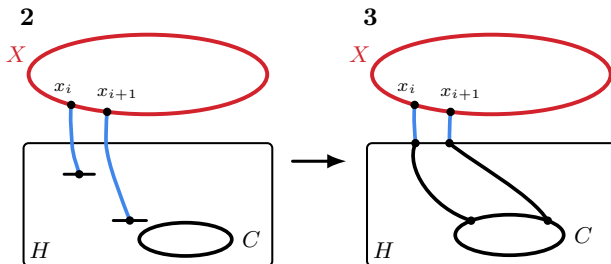
$$c(G) \geq k \Omega(c(G)/m) \implies m = \Omega(k).$$

Two key properties in the proof

In general, the components of $G - X$ need not be 3-connected. It suffices to recover the following two properties.

Property 1

Two disjoint (X, H) -paths can be rerouted within H to end on a common cycle.



This requires a weak form of 2-connectivity.

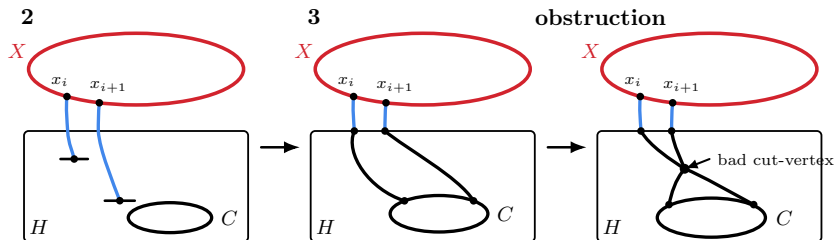
Bad cut-vertex: obstruction to Property 1

Let X be a longest cycle, and let H be a component of $G - X$.

The obstruction to Property 1 lies in a certain type of cut-vertices in H .

Property 1

Two disjoint (X, H) -paths can be rerouted within H to end on one common cycle.



Weak 2-connectivity: it suffices to eliminate the cut-vertices that separate the paths in $Y - X$.

Achieve Property 1: a weakly 2-connected covering

To achieve Property 1, we repeatedly split the graph H to eliminate these bad cut-vertices. We record some features during this process:

$\mathcal{P}$	$\mathcal{H}$	$\mathcal{S}$
Segments paths generated by $Y - X$	Envelopes edge-disjoint subgraphs containing the segments	Recorded cuts bad cut-vertices so far

Start from the paths of $Y - X$; update all three objects after every split.

The covering theorem

Theorem

Let $\mathcal{P}_0$ be the nontrivial path components of $Y - X$. There exists a triple system $(\mathcal{P}, \mathcal{H}, S)$ such that:

- $E(\mathcal{P}) = E(\mathcal{P}_0)$, $|\mathcal{H}| \leq |\mathcal{P}| \leq 6m$, $|S| \leq 6m$, and no envelope has a bad cut-vertex;
- if a cycle $C \subseteq H$ meets the segments in H and two disjoint paths from X enter H at x_1, x_2 , then

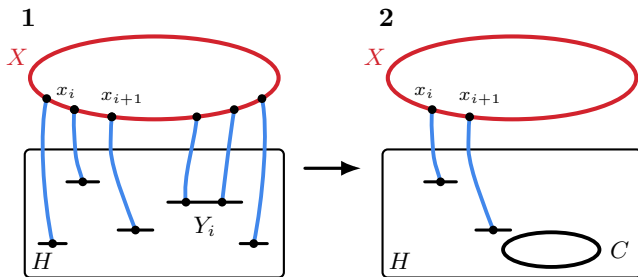
$$\text{dist}_X(x_1, x_2) \geq |C|/2.$$

- Main difficulty: the final numbers of envelopes and segments remain $O(m)$.
- Proof: repeatedly apply a graph split operation, ending with the desired subgraphs as the envelopes.

Amplifiers

Property 2

For some long path P in H , find a cycle $C \subseteq H$ with length $\Omega(|P|)$.



We use **amplifiers** to describe the paths with Property 2.

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Definition $((H, \alpha)$ -amplifier)

A path $P \subseteq H$ is an (H, α) -amplifier if there is a cycle $C \subseteq H$ such that

$$|E(C) \cap E(P)| \geq \alpha |E(P)|.$$

Theorem (Bondy–Locke)

Every 3-connected graph H satisfies $c(H) \geq \frac{2}{5}p(H)$.

Their proof actually implies: every path in a 3-connected graph H is an $(H, 2/5)$ -amplifier.

Cover the paths by amplifiers and leftovers

Here $\mathcal{P}$ consists of the path segments obtained from $Y - X$ after splitting.

Lemma 5.1

There exists a collection $\mathcal{U}$ of subpaths of $\mathcal{P}$ such that:

- every $U \in \mathcal{U}$ is a $(G - X, 1/6)$ -amplifier;
- $|V(\mathcal{P}) \setminus V(\mathcal{U})| = O(mk)$.

Fix a segment P in an envelope H . Every component K_i of $H - P$ determines an interval $P[u_i, v_i]$, which is covered by at most three amplifiers.

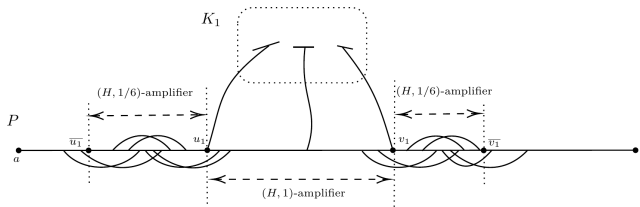


Figure 5: Covering the interval $P[\overline{u_1}, \overline{v_1}]$ by three amplifiers

Repeat this construction for every segment in $\mathcal{P}$.

Cover the paths by amplifiers and leftovers

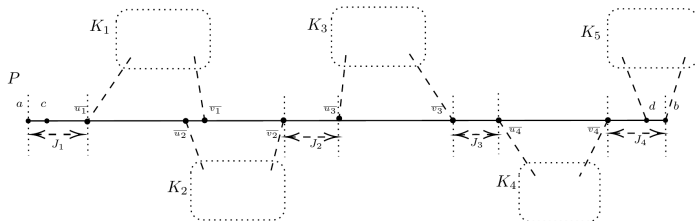
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The intervals not covered by any component of $H - P$ are separated from the other prescribed paths by adding more vertex cuts to $\mathcal{S}$.



Consequently, the edges of P are partitioned into amplifier subpaths and some short leftovers.

The remaining range

Theorem (Ma–Ning–Z.)

For every integer $k \geq 2$, any two longest cycles in a k -connected graph share at least $k/4000$ vertices.

The previous ideas already prove the main theorem when $c(G) > 100mk$. It remains to consider $c(G) \leq 100mk$, where the previous covering idea may fail, as the resulting $O(m)$ segments may all have length $O(k)$.

We use an independent tool: a conjecture due to Jung (1990) that strengthens Dirac's theorem (1952).

Dirac's theorem to Jung's conjecture

The following result is classical.

Theorem (Dirac, 1952)

Every 2-connected n -vertex graph with minimum degree δ contains a cycle of length at least

$$\min\{2\delta, n\}.$$

Motivated by this, Jung conjectured the coefficient 2 would increase with connectivity.

Conjecture (Jung, 1990/2001)

Let G be a k -connected graph with minimum degree δ , and let C be a longest cycle in G . If $G - V(C)$ contains a path on at least $k - 1$ vertices, then

$$|C| \geq k(\delta - k + 2).$$

Jung's conjecture holds when $\delta \geq 6k$

We proved Jung's conjecture when $\delta \geq 6k$.

Theorem (Ma–Ning–Z., 2026, preprint)

Let G be a k -connected graph with minimum degree $\delta \geq 6k$, and let C be a longest cycle in G . If $G - V(C)$ contains a path on at least $k - 1$ vertices, then

$$|C| \geq k(\delta - k + 2).$$

Application to the remaining case. Suppose $c(G) \leq 100mk$ and $k > 4000m$.

Verifying the parameters: G is $c(G)/(600m)$ -connected,

$$\delta(G) \geq k \geq 6 \frac{c(G)}{600m}, \quad G - X \text{ contains a path on at least } \frac{c(G)}{600m} - 1 \text{ vertices.}$$

Applying MNZ gives

$$c(G) \geq \frac{c(G)}{600m} \left(k - \frac{c(G)}{600m} \right) > c(G),$$

a contradiction. Hence $m \geq k/4000$.

Ma–Ning–Zhao, arXiv:2606.03696 (2026)

Three independent ingredients

In brief, we establish a linear form of Smith's conjecture, using the following ingredients.

- **Achieving Property 1:** a weak-2-connectivity covering theorem.
- **Achieving Property 2:** a generalization of Bondy–Locke's result.
- **Remaining case:** a Dirac-type theorem for highly connected graphs.

We hope that these independent results are useful in other problems.

Thank you!